CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

May 2012

CL101 Fundamentals of Civil Engineering

Date: 11.05.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define civil engineering and infiltration rate. [02]

(b) Explain role of civil engineers. [05]

Q - 2 (a) Describe orientation of building and site selection in detail. [04]

(b) Classify the building based on structures. [05]

(c) Explain hydrological cycle with neat sketch. [05]

OR

Q - 2 (a) Enlist principles of building planning and explain any two in detail. [06]

(b) Give the classification of foundation. [03]

(c) Write a note on methods of conservation of water. [05]

Q - 3 (a) Give the classification of canal. [05]

(b) Write the functions of foundation, column, plinth, window and wall. [05]

(c) Enlist different types of loads and explain any one in detail. [04]

OR

Q - 3 (a) Define hydraulic structures and explain gravity dam in detail. [05]

(b) Draw the detailed plan of a building shown in **Figure 1.** Also give the dimensions of door, window, floor height, plinth, riser, tread and show the same in detailed plan at appropriate locations. [09]

SECTION – II

Q - 4 (a) Write conversion of units for following: [02]

1 hector = _______ m^2.

1 foot = _______ m.

(b) Enlist instruments used for laying offsets. [02]

(c) Sketch conventional symbols for following: [03]

Canal, Railway line (double), Traverse station

Q - 5 (a) State fundamentals principles of surveying. [02]

(b) Convert the bearings: [02]

(i) 238° & (ii) N 45° W

(c) State uses of theodolite. [03]

(d) Calculate the interior angles of the traverse ABCDE from following bearings. Also apply usual checks. [07]

Line	F.B.	B.B.
AB	107° 00’	287° 00’
BC	22° 00’	202° 00’
CD	281° 30’	101° 30’
DE	189° 00’	9° 00’
EA	124° 30’	304° 30’

OR

Q - 5 (a) Explain the following terms: **[02]**
Line of collimation and Reduced level

(b) Magnetic bearing of a line is 200°. Calculate the true bearing if the magnetic declination is 4° W. **[02]**

(c) Enlist different types of level. **[03]**

(d) The following staff readings were observed successively with a level, the instruments having been moved after third, sixth and eight readings. 2.225, 1.605, 0.995, 2.090, 2.865, 1.265, 0.600, 1.985, 1.045, 2.685 m. Enter the above readings in a page of level book and calculate the reduced levels of all points If the first reading was taken with a staff held on bench mark of 135.75 m. Use H.I. method. **[07]**

Q - 6 (a) Write a note on errors in chaining. **[03]**

(b) Discuss suitability and requirements of different transportation modes. **[07]**

(c) A 20 m chain was found to be 10 cm too long after chaining a distance of 1500m. It was found to be 18 cm too long at the end of one day’s work after chaining the total distance of 3900 m. Find the true distance if chain was correct before the commencement of work. **[04]**

OR

Q - 6 (a) Differentiate between prismatic compass and surveyor’s compass. **[04]**

(b) Briefly describe components of G.I.S. **[03]**

(c) Write a note on B.O.T. projects for highway. **[07]**

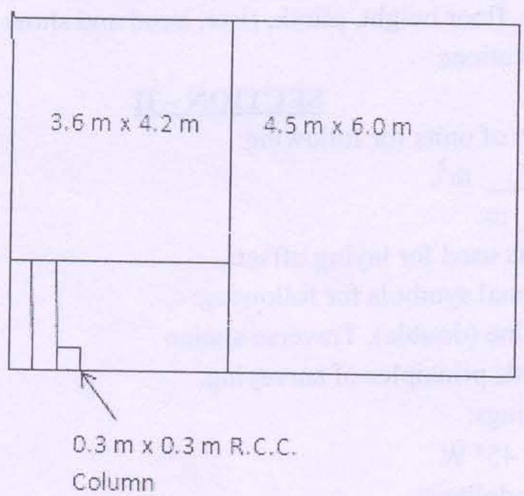

Figure 1